8. Torsion of Circular Shafts

8.1 Introduction

When equal and opposite forces are applied tangentially to the ends of a shaft it is subject to a twisting moment which is equal to the product of force applied and radius of shaft. This causes shaft to rotate with constant angular velocity. A shaft is said to be in pure torsion when it is subjected to equal and opposite torques at its two ends. It results in shearing off of shaft at every cross section perpendicular to its longitudinal axis. While a beam bends as an effect of bending moment, a shaft twists as an effect of torsion.

8.2 Theory of pure torsion

Following assumption are made, while finding out shear stress in a circular shaft subjected to torsion:

1. No distortion of parallel planes normal to axis takes place.
2. Torsion is uniform along the rod i.e. all the normal cross sections which are at the same axial distance suffer equal relative rotation.
3. All the stresses are within the elastic limit.
4. The material is homogenous and isotropic.
5. The shaft is circular in section remains circular after twisting.

Let one end of the shaft is fixed and a moment is applied at the other end. This is valid because whether it rotates at uniform speed to transmit the power or is at rest, the stress and strains due to equal and opposite couples at its ends will remain the same.

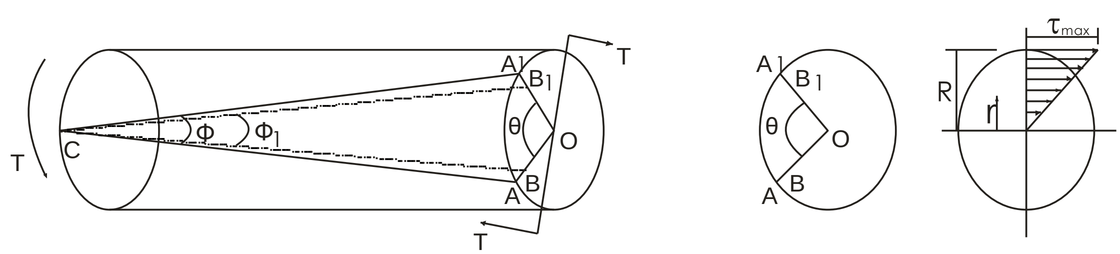

Fig. 8.1

Let us consider a torque T applied at the free end of the shaft as in Fig 8.1. A balancing torque of equal magnitude and opposite in direction will be induced at the fixed end. Let the cross section be distorted through an angle θ, shifting the radii OA to OA_1. On the surface of the shaft any line such as CA will be distorted to CA_1, through an angle ϕ. The angle L ACA' = ϕ is called the shear stain and $\angle$BOB' = θ is called the angle of twist.

We know that shear strain = deformation per unit length = $\frac{AA_1}{l} = \tan\phi = \phi$

(ϕ being small $\therefore$ $\tan\phi = \phi$)

If τ be the intensity of shear stress at the surface of the shaft and C be the modulus of rigidity or shear modulus:

$$\text{Shear strain} \quad \phi = \frac{\tau}{C} \quad [\because \text{ shear stress} \propto \text{shear strain}] \qquad \text{.....(8.1)}$$

$$\text{Also} \quad \phi = \frac{AA'}{CA} = \frac{OA.\theta}{CA} = \frac{R\,\theta}{l} \qquad \text{.....(8.2)}$$

$$\because \quad \phi = \frac{\tau}{C} = \frac{R\,\theta}{l} \qquad \text{.....(8.3)}$$

$$\frac{\tau}{R} = \frac{C\,\theta}{l} \qquad \text{..... (8.4)}$$

Similarly a layer such as DB distant r from the axis of the shaft will be distorted to a new position DB_1 through an angle ϕ_1 and we have

$$\frac{q}{C} = \phi_1 = \frac{BB_1}{DB_1} = \frac{r\,\theta}{l}$$

$$\frac{q}{r} = \frac{C\,\theta}{l} \qquad \text{.....(8.5)}$$

$$\text{Hence} \quad \frac{\tau}{R} = \frac{q}{r} = \frac{C\,\theta}{l} \qquad \text{..... (8.6)}$$

Intensity of shear stress at any point in the cross section of a shaft subject to pure torsion is proportional to its distance from the centre. This shear intensity is zero at axis and increases linearly to maximum of τ at surface as shown in Fig. 8.1.

Resisting Torque:

To calculate the torque, a shaft can withstand or transmit, let us consider an elementary ring of radius r and thickness δr. Let the shear intensity of stress at this radius be q .we have

$$q = \frac{\tau}{R} r \qquad \text{(from equation 8.6)}$$

the resistance couple set up by the ring

$$\delta T_r = q.\delta a.r$$

(δa = area of the ring = $2\pi r\delta r$)

$$\delta T_r = \frac{\tau}{R}.r.\delta a.r$$

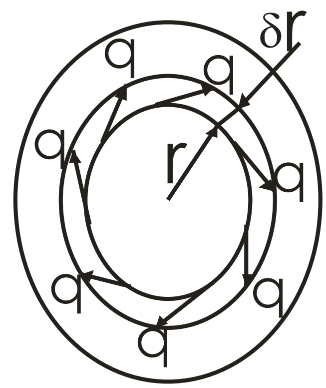

Fig. 8.2

Resistance by the whole shaft

$$T_r = \sum_O^R \delta T_r = \int_O^R \frac{\tau}{R}\delta a r^2 = \frac{\tau}{R}\int_O^R \delta a\, r^2$$

$\int_O^R \delta a r^2$ represents the polar moment of inertia J about the longitudinal axis of the solid shaft. The moment of inertia of a plane area w.r.t. an axis perpendicular to the plane of the figure is called polar moment of inertia or w.r.t. the point where the axis intersects the plane.

Hence $\quad T_r = \frac{\tau}{R}J$

$$\frac{T_r}{J} = \frac{\tau}{R} \qquad \text{.....(8.7)}$$

Combining equations (8.6) & (8.7) we have torsion equation

$$\frac{q}{r} = \frac{\tau}{R} = \frac{T_r}{J} = \frac{C\theta}{l}$$

Note: for bending $\quad \frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$

In a beam the bending moment produces a bend or deflection, in the same manner a torque produces a twist in a shaft. The expression CJ corresponds to a similar quantity EI in expression for deflections of beams. The quantity CJ is called torsional rigidity while EI is called flexural rigidity.

8.3 Power Transmitted By A Shaft

Let a shaft turning at N rpm transmits P KW. Let the mean torque to which the shaft is subjected to be T Nm.

Power transmitted P = mean torque x angle turned / sec.

$$P = T \times \frac{N}{60} \times 2\pi \quad \text{W}$$

$$P = \frac{2\pi N T}{60} W \qquad \text{.....(8.8)}$$

8.6 Composite Circular Shafts

8.6.1 Shafts in Series:

Consider a shaft made of two material connected in series as shown in Fig. 8.4. The shaft is subjected to a torque T. As both the shafts are subjected to the same torque

$$\therefore \qquad T_1 = T_2 \qquad \text{.....(8.13)}$$

Now $\frac{T_1}{J_1} = \frac{\tau_{S1}}{\frac{D_1}{2}} = \frac{C_1\theta_1}{L_1}$ and

$$\frac{T_2}{J_2} = \frac{\tau_{S2}}{\frac{D_2}{2}} = \frac{C_2\theta_2}{L_2}$$

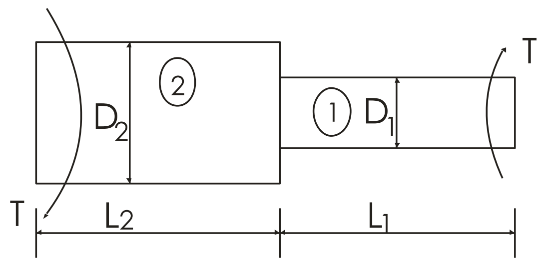

Fig. 8.4

Total angle of twist, $\theta = \theta_1 + \theta_2$

8.6.2 Shafts in Parallel:

Now consider a shaft made of two material connected in parallel as shown in Fig. 8.5. The torque applied in this case is divided between the two shafts but the angle of twist is the same for each shaft. Let T be the applied torque.

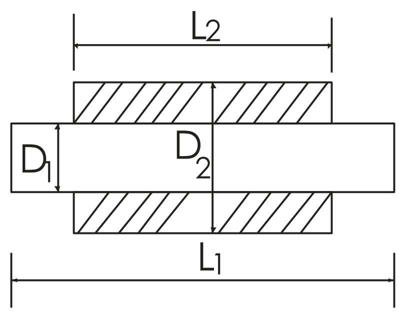

Fig. 8.5

Then $T = T_1 + T_2$ and $\theta_1 = \theta_2$.

8.5 Comparison Between Hollow And Solid Shaft

Let D_1 = Diameter of solid shaft

L = length of shafts

ρ = density of metal

D = Outer diameter of hollow shaft

d = inner diameter of hollow shaft

n = D/d

Polar moment of inertia of solid shaft = $\dfrac{\pi D_1^4}{32}$

Polar moment of inertia of hollow shaft = $\dfrac{\pi(D^4 - d^4)}{32}$

Comparison can be done on two conditions:

(a) **By Strength**: In this case we assume both solid and hollow shafts to be of same length, weight and material and we find the ratio of toque transmitted by both the shafts:

Using torsion equation $\dfrac{T}{J} = \dfrac{\tau}{R}$

$$\frac{T_{hollow}}{T_{solid}} = \frac{J_{hollow} R_{solid}}{R_{hollow} J_{solid}} = \frac{D^4 - d^4}{D_1^4} \times \frac{D_1}{D} = \frac{d^3 (n^4 - 1)}{n D_1^3} \qquad \text{.....(8.9)}$$

Since weight of both the shafts are same

Hence $\dfrac{\pi}{4} D_1^4\, l\, P = \dfrac{\pi}{4} (D^2 - d^2)\, l\, P$

or $D_1 = \sqrt{D^2 - d^2} = d\sqrt{n^2 - 1}$

or $D_1^3 = d^3 (n^2 - 1)\sqrt{n^2 - 1}$ .....(8.10)

putting this into equation (8.9)

$$\frac{T_{hollow}}{T_{solid}} = \frac{n^4 - 1}{n\sqrt{n^2 - 1}\,(n^2 - 1)} = \frac{n^2 + 1}{n\sqrt{n^2 - 1}} \qquad \text{.....(8.11)}$$

Since n >1 always hence hollow shaft is stronger than the solid shaft for same metal and weight.

If n = 2

$$\frac{T_{hollow}}{T_{solid}} = \frac{4 + 1}{2\sqrt{4 - 1}} = 1.44$$

(b) **Comparison By Weight** : In this case for the same torque transmitted the weight ratio of hollow and solid shafts are found out

$$\frac{W_{hollow}}{W_{solid}} = \frac{\pi(D^2 - d^2) R \times density}{\pi D_1^2\, r\, l \times density} = \frac{D^2 - d^2}{D_1^2}$$

$$= \frac{d^2(n^2-1)}{D_1^2} \qquad (D = nd)$$

Since the Torque transmitted are same

$$T_{hllow} = T_{solid}$$

$$\frac{\tau}{R} \times \frac{\pi}{32}(D^4 - d^4) = \tau_s \times \frac{\pi}{32} \times D_1^3$$

$$\frac{D^4 - d^4}{D} = D_1^3$$

$$\frac{d^4(n^4-1)}{D} = D_1^3$$

$$\frac{d^3(n^4-1)}{n} = D_1^3$$

$$D_1^2 = d^2\left(\frac{n^4-1}{n}\right)^{2/3}$$

$$\frac{W_{hollow}}{W_{solid}} = \frac{(n^2-1)}{\left(\frac{n^4-1}{n}\right)^{2/3}} = \frac{n^{2/3}(n^2-1)}{(n^4-1)^{2/3}} \qquad \text{.....(8.12)}$$

If $n = 2$

$$\frac{W_{hollow}}{W_{solid}} = 0.7829$$

For a given torque weight of the hollow shaft will be lesser than solid shaft.